

Revised Generation Interconnection System Impact Study Report

For Queue Project Q20-1619

Blue Ravine-School 230 kV
January 20, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1619. The Interconnection Customer proposes a 157.6 MW solar photovoltaic facility interconnecting at the Blue Ravine-School 230 kV substation in the Indiana area. The request was received on December 08, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 563 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,984,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Myrtle Pt - George Neal S 230 kV	Swan Vly - Trego Substation 500 kV	127.5	1,448	8.0
George Neal South Substation - Zona Libre 6007 Substation 230 kV	Swan Valley - Trego 500 kV	126.9	1,476	8.0
Montpelier Electric Generating Sta - Dooville 230 kV	Montpelier Electric Generating Station - Wabash 500 kV	113.3	1,172	14.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Myrtle Pt - George Neal S 230 kV	Rebuild line with higher-capacity conductor	\$21,326,000	2% (\$524,000)
George Neal South Substation - Zona Libre 6007 Substation 230 kV	Rebuild line with higher-capacity conductor	\$34,870,000	2% (\$857,000)
Dooville - Montpelier Electric Generating Sta 230 kV	Rebuild line with higher-capacity conductor	\$24,198,000	2% (\$603,000)
	Total allocated to project		\$1,984,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.